

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Sundaresan.R.U	Reg. No.:	192424359
Section / Batch:	2024	Date:	24/07/2026

Code of Conduct

I Sundaresan.R.U/192424359 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sundaresan.R.U

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1 Binary Search

5 marks

- Q1**
Binary Search
5 marks
- Q2**
Stack
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Stack - Balancing Symbols
5 marks
- Q6**
Singly Linked List
5 marks
- Q7**
Stack
5 marks
- Q8**
Linked List
5 marks
- Q9**
Linked List
5 marks
- Q10**
Queue
5 marks

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while (low < high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 int binarySearch(int arr[], int n, int key) {
2     int low = 0, high = n - 1;
3
4     while (low <= high) {
5         int mid = low + (high - low) / 2;
6
7         if (arr[mid] == key)
8             return mid;
9         else if (arr[mid] < key)
10            low = mid + 1;
11        else
12            high = mid - 1;
13    }
14
15    return -1;
16 }

```

Test Input

```
stdin input (optional)
```

Output

```
Compilation Error:
C:\CCPPComp\CCPP2\bin\...\lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
C:\CCPPComp\CCPP2\bin\...\lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
C:\CCPPComp\CCPP2\bin\...\lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
C:\CCPPComp\CCPP2\bin\...\lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
```

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← Back

Q2 Stack

5 marks


```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test Input

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7 void push(int x) {
8     if (top == MAX - 1) {
9         printf("Overflow\n");
10        return;
11    }
12    stack[++top] = x;
13 }
14
15 int pop() {
16     if (top == -1) {
17         printf("Underflow\n");
18         return -1;
19     }
```

```
stdin input (optional)
```

Output

```
Test 1: pop on empty stack
Underflow
Returned: -1

Test 2: push 10,20,30 then pop
three
```

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Q3 Stack

5 marks

Q10

Queue

5 marks

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stackArr[MAX];
6 int top;
7
8 void init() {
9     top = -1; // empty stack
10 }
11
12 int isEmpty() {
13     return (top == -1);
14 }
15
16 int isFull() {
17     return (top == MAX - 1);
18 }
19
20 void push(int x) {
21     if (isFull()) {
22         printf("Overflow! Cannot push %d\n", x);
23         return;
24     }
25     stackArr[++top] = x;
26 }
```

Test Input

```
stdin input (optional)
```

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q4 Stack 5 marks

Identify the errors in the following snippet.

```
if(ch == 'J') {  
    while(stack[top] != 'J') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3 #include <string.h>  
4  
5 #define MAX 100  
6  
7 char stack[MAX];  
8 int top = -1;  
9  
10 void push(char x) {  
11     stack[++top] = x;  
12 }  
13  
14 char pop() {  
15     if (top == -1) return '\0';  
16     return stack[top--];  
17 }  
18  
19 int isEmpty() {  
20     return top == -1;  
21 }  
22  
23 int precedence(char op) {  
24     if (op == '^') return 3;
```

Test Input

stdin input (optional)

Output

Security Error: Disallowed code pattern detected.

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5 marks

Q10 

Queue

5 marks

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
    pop();
} else {
    return 0;
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test Input

stdin input (optional)

Output

```
Test 1: "{[()]}"
Expected: 1, Got: 1 => PASS

Test 2: "{[]}"
Expected: 1, Got: 1 => PASS
```

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```

1 #include <stdio.h>
2 #include <string.h>
3
4 int isBalanced(const char *expr) {
5     int n = (int)strlen(expr);
6     char stack[n];
7     int top = -1;
8
9     for (int i = 0; i < n; i++) {
10         char ch = expr[i];
11
12         if (ch == '(' || ch == '[' || ch == '{') {
13             stack[++top] = ch;
14         }
15         else if (ch == ')' || ch == ']' || ch == '}') {
16             if (top == -1) return 0;
17             if (ch == ')' && stack[top] == '(') ||
18                 (ch == ']' && stack[top] == '[') ||
19                 (ch == '}' && stack[top] == '{') {
20                 top--;
21             } else {
22                 return 0;
23             }
24         }
25     }
26     return 1;
27 }

```

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

Q6

Singly Linked List

5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct Node {
5     int data;
6     struct Node* next;
7 } Node;
8
9 Node* createNode(int val) {
10     Node* n = (Node*)malloc(sizeof(Node));
11     n->data = val;
12     n->next = NULL;
13     return n;
14 }
15
16 void printList(Node* head) {
17     Node* temp = head;
18     while (temp != NULL) {
19         printf("%d", temp->data);
20         if (temp->next != NULL) printf(" ");
21         temp = temp->next;
22     }
23 }
24
25 int main() {
26     int n;
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0;

print("Underflow")

else:

return stack[top--]

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 void PUSH(int val) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11     } else {
12         stack[++top] = val;
13     }
14 }
15
16 int POP() {
17     if (top == -1) {
18         printf("Underflow\n");
19         return -1;
20     } else {
21         return stack[top--];
22     }
23 }
24
25 int main() {
```

Test Input

10 20 30

Output

Popped: 30

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

Q8

Linked List

5 marks

Identify the errors in the following code

while current != NULL:
current.next = prev
prev = current
current = current.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 unsigned long long factorial(int n) {
3     if (n < 0) {
4         return 0;
5     }
6     unsigned long long fact = 1;
7     for (int i = 1; i <= n; i++) {
8         fact *= i;
9     }
10    return fact;
11 }
12
13 int main() {
14     int num;
15
16     printf("Enter a positive integer: ");
17     if (scanf("%d", &num) != 1) {
18         printf("Invalid input. Please enter an integer.\n");
19         return 1;
20     }
21
22     if (num < 0) {
23         printf("Factorial is not defined for negative numbers.\n");
24     } else {
25         printf("Factorial of %d = %llu\n", num, factorial(num));
26     }
27 }
```

Test Input

1 2 3 4 5

Output

Enter a positive integer: Factorial
of 1 = 1

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q9 Linked List 5 marks

What is the error in the below code?
new.next = head
head = new

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct Node {
5     int data;
6     struct Node* next;
7 } Node;
8
9 Node* newNode(int x) {
10     Node* n = (Node*)malloc(sizeof(Node));
11     n->data = x;
12     n->next = NULL;
13     return n;
14 }
15
16 void insertAtHead(Node** head, int x) {
17     Node* new = newNode(x);
18     new->next = *head;
19     *head = new;
20 }
21
22 void printList(Node* head) {
23     while (head != NULL) {
24         printf("%d ", head->data);
25         head = head->next;
26     }
27     printf("\n");
28 }
29
```

Test Input

48 50 60 70

Output

70 40 50 60

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q10 Queue5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

🚩 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int Q[MAX];
2 int front = -1;
3 int rear = -1;
4
5 /
6 int dequeue() {
7     if (front == -1 || front > rear) {
8         printf("Underflow\n");
9         return -1;
10    } else {
11        int element = Q[front];
12        front++;
13
14        if (front > rear) {
15            front = rear = -1;
16        }
17        return element;
18    }
19 }
20
21 int main() {
22
23     Q[0] = 10;
24     Q[1] = 20;
25     Q[2] = 30;
```

Test Input

10 20 30

Output

10 20 30

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Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Point out the error?
PEEK(stack);
return stack[top+1]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int stack[MAX];
5 int top = -1;
6 void push(int value) {
7     if (top == MAX - 1) {
8         printf("Stack Overflow\n");
9         return;
10    }
11    stack[++top] = value;
12 }
13 int peek() {
14     if (top == -1) {
15         printf("Stack is empty\n");
16         return -1;
17     }
18     return stack[top];
19 }
20
21 int main() {
22     push(10);
23     push(20);
24     push(30);
25
26     printf("Top element is: %d\n", peek());
27
28     return 0;
29 }
```

Test Input

stdin input (optional)

Output

Top element is: 30

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- 10/10 answered

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3
4 int queue[MAX];
5 int front = -1, rear = -1;
6
7 void enqueue(int vehicle) {
8     if ((rear + 1) % MAX == front) {
9         printf("Queue Full\n");
10    } else {
11        if (front == -1) {
12            front = 0;
13        }
14        rear = (rear + 1) % MAX;
15        queue[rear] = vehicle;
16        printf("Vehicle %d added to queue\n", vehicle);
17    }
18 }
19
20 void dequeue() {
21     if (front == -1) {
22         printf("Queue Empty\n");
23     } else {
24         printf("Vehicle %d removed from queue\n", queue[front]);
25         if (front == rear) {
26             front = -1;
27             rear = -1;
28         }
29     }
30 }

```

stdin input (optional)

```
Vehicle 101 added to queue
Vehicle 102 added to queue
Vehicle 103 added to queue
Vehicle 104 added to queue
Vehicle 105 added to queue
Queue Full
```

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- Point out the error.
if top == MAX:
Overflow

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 5
5
6 int stack[MAX];
7 int top = -1;
8
9 void push(int value) {
10     if (top == MAX - 1) {
11         printf("Overflow\n");
12         return;
13     }
14     top++;
15     stack[top] = value;
16 }
17
18 int main() {
19     push(10);
20     push(20);
21     push(30);
22     push(40);
23     push(50);
24     push(60);
25
26     return 0;
27 }

```

Test Input

stdin input (optional)

Output

Overflow

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10/10 answered

Submitted

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Q1

Stack

5 marks

Q2

Circular Queue

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int x) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13     stack[++top] = x;
14 }
15
16 int pop() {
17     if (top == -1) {
18         printf("Underflow\n");
19         return -1;
20     }
21     int item = stack[top--];
22     return item;
23 }
24
25 int main() {
26     int i, val;
27
28     val = pop();
29     printf("Popped: %d\n", val);
30 }
31
```

Test Input

stdin input (optional)

Output

Underflow
Popped: -1
Popped: 30
Popped: 20
Popped: 10
Underflow

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Q2

Circular Queue

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* prev;
7     struct Node* next;
8 };
9
10 void deleteNode(struct Node** head_ref, struct Node* del) {
11     if (*head_ref == NULL || del == NULL) return;
12
13     if (*head_ref == del) {
14         *head_ref = del->next;
15     }
16
17     if (del->next != NULL) {
18         del->next->prev = del->prev;
19     }
20
21     if (del->prev != NULL) {
22         del->prev->next = del->next;
23     }
24
25     free(del);
26 }
27
28 void push(struct Node** head_ref, int new_data) {
29     struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
30     new_node->data = new_data;
```

Test Input

stdin input (optional)

Output

Original List: 40 30 20 10
List after deleting 30: 40 20 10

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Sundaresan.R.U

192424359RA

Q2

Circular Queue

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Doubly Linked List

5 marks

✓

Q6

Queue

5 marks

✓

Q7

Queue - Deque

5 marks

✓

Q8

Binary Search

5 marks

✓

Q9

Circular Queue

5 marks

✓

Q10

Stack - Postfix Expression

5 marks

✓

10/10 answered

queue[rear++] = data;

}

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 int isFull() {
9     return (rear == MAX - 1);
10 }
11
12 int isEmpty() {
13     return (front == -1 || front > rear);
14 }
15
16 void enqueue(int data) {
17     if (isFull()) return;
18     if (front == -1) front = 0;
19     queue[++rear] = data;
20 }
21
22 int dequeue() {
23     if (isEmpty()) return -1;
24     int item = queue[front];
25     front++;
26     if (front > rear) {
27         front = -1;
28         rear = -1;
29     }
30     return item;
31 }
32
```

Test Input

stdin input (optional)

Output

10 20 30 40 50

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Stack - Postfix Expression

5 marks

10/10 answered

else front = front - 1;
deque[front] = data;
}

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int dequeArr[MAX];
6 int front = -1, rear = -1;
7
8 int isFull() {
9     return (front == 0 && rear == MAX - 1) || (front == rear + 1);
10 }
11
12 int isEmpty() {
13     return front == -1;
14 }
15
16 void insertFront(int data) {
17     if (isFull()) {
18         printf("Deque Overflow\n");
19         return;
20     }
21     if (isEmpty()) {
22         front = rear = 0;
23     } else if (front == 0) {
24         front = MAX - 1;
25     } else {
26         front = front - 1;
27     }
28     dequeArr[front] = data;
29 }
30
31 void insertRear(int data) {
```

Test Input

stdin input (optional)

Output

Deque elements: 40 20 10 30
Deque elements: 20 10 30
Deque elements: 50 20 10 30 60
Deque Overflow
Deque elements: 50 20 10 30 60

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5 marks

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Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

if (arr[mid] == x) return mid;

if (arr[mid] > x) return binarySearch(arr, l, mid, x);

return binarySearch(arr, mid, r, x);

}

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (r >= 1) {
5         int mid = l + (r - l) / 2;
6
7         if (arr[mid] == x)
8             return mid;
9
10        if (arr[mid] > x)
11            return binarySearch(arr, l, mid - 1, x);
12
13        return binarySearch(arr, mid + 1, r, x);
14    }
15    return -1;
16 }
17
18 int main() {
19     int arr[] = {1, 3, 5, 7, 9, 11, 13, 15};
20     int n = sizeof(arr) / sizeof(arr[0]);
21
22     int x = 11; // test value
23
24     int result = binarySearch(arr, 0, n - 1, x);
25
26     if (result != -1)
27         printf("Element %d found at index %d\n", x, result);
28     else
29         printf("Element %d not found\n", x);
30 }
```

Test Input

stdin input (optional)

Output

Element 11 found at index 5

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School



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Circular Queue

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Doubly Linked List

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5 marks

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Binary Search

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Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

int data = queue[front];

front = (front + 1) % MAX;

return data;

}

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queueArr[MAX];
6 int front = -1;
7 int rear = -1;
8
9 int enqueue(int x) {
10     if ((rear + 1) % MAX == front) return -1; // full
11     if (front == -1) front = 0;
12     rear = (rear + 1) % MAX;
13     queueArr[rear] = x;
14     return 0;
15 }
16
17 int dequeue() {
18     if (front == -1) return -1; // empty
19
20     int data = queueArr[front];
21
22     if (front == rear) {
23         front = rear = -1;
24     } else {
25         front = (front + 1) % MAX;
26     }
27     return data;
28 }
29
30 int main() {
```

Test Input

stdin input (optional)

Output

-1
10
20
30
40
50

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- 10/10 answered

```

else {
    int op1 = pop();
    int op2 = pop();
    // perform operation: result = op1 + op2;
    push(result);
}
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <ctype.h>
3
4 #define MAX 100
5
6 int stack[MAX];
7 int top = -1;
8
9 void push(int x) {
10     if (top < MAX - 1) stack[++top] = x;
11 }
12
13 int pop() {
14     if (top < 0) return 0;
15     return stack[top--];
16 }
17
18 int evalPostfix(char expr[]) {
19     int i = 0;
20
21     while (expr[i] != '\0') {
22         if (isspace((unsigned char)expr[i])) {
23             i++;
24             continue;
25         }
26     }

```

Test Input

stdin input (optional)

Output

```
Security Error: Disallowed code
pattern detected.
```

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